MILITARY SPECIFICATION

SIGHT, INFINITY: 8635466

1. SCOPE

1.1 This specification covers Sight, Infinity: 8635466, utilized to project a reticle into the field of a gunner's periscope and control the light intensity of the projected reticle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–O–13830 —Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of

MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

MIL–P–14232 —Parts, Equipment, and Tools for Ordnance Materiel; Packaging of

MIL–I–45208 —Inspection Requirements, General Specification for

MIL–I–45607 —Inspection Equipment, Supply, and Maintenance for Ordnance

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedure and Tables for Inspection by Attributes

DRAWINGS

MUNITIONS COMMAND

F8635466 —Sight, Infinity: 8635466

IEL 8635466 —Index of Inspection Equipment List for Sight, Infinity: 8635466

LSQAP 8635466 —Supplementary Quality Assurance Provision List for Sight, Infinity: 8635466

PACKAGING DATA SHEET

MIL–P–14232/P8635466 —Packaging of Sight, Infinity: 8635466

(Copies of Specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 **Materials.** The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.3 **Design and construction.** The sight shall be manufactured in accordance with Drawing

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F8635466, "Sight, Infinity: 8635466" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product.

3.4 **General specification.** The sight shall meet the following requirements of Specifications MIL–O–13830 and MIL–F–13926:

MIL–O–13830

(a) Cleanliness.
(b) Fixed eyepiece focus.

MIL–F–13926

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Electrical and electronic assemblies.
(f) Workmanship.

3.5 **Alignment.** With the sight positioned as shown on Drawing F8635466 in a vertical plane and the locating key horizontal within plus or minus 1 minute, the line of sight from the center of the reticle circle shall be 19 degrees below the plane of the locating key as adjusted by the boresight knobs.

3.5.1 *Knob travel.* With the line of sight positioned as above, the boresight knobs shall provide at least 10 mils movement in both directions in azimuth and elevation.

3.6.1 *Storage temperature.* The sight shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of +15° C (+60° F.) to 32° C. (+90° F.) after having been exposed and thermally stabilized at ambient temperatures of +71.1° C. (+160° F.) and −62.2° C. (−80° F.).

3.6.2 *Vibration.* The sight shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration, there shall be no evidence of loose or damaged parts.

3.6.2.1 *Reticle retention.* After being subjected to the requirements of 3.6.1 and 3.6.2 the line of sight shall not have changed more than 0.5 mil from the setting established in 3.5 and the sight shall meet the requirements of 3.6.3 to 3.7.5 inclusive.

3.6.3 *Operating temperatures.* Subsequent to being subjected to temperatures of +65.6° C. (+150° F.) and −40° C. (−40° F.) until thermal equilibrium is reached the sight shall meet the requirements of 3.7.4.

3.6.4 *Sealing.* The sight shall show no evidence of leakage when subjected to an external pressure of 5 psi (gage pressure) for a period of five minutes.

3.7 **Performance.**

3.7.1 *Illumination.* Utilizing a 24 volt source to illuminate the reticle, the reticle pattern shall appear uniformly illuminated and free of ragged edges or breaks throughout its entire circumference.

3.7.2 *Focus.* The reticle shall be in focus between zero and minus 1/8 diopters.

3.7.3 *Reticle.* The inner diameter of the reticle circle shall subtend an angle of 21 mils plus or minus 3 mils.

3.7.4 *Torque.*

3.7.4.1 *Boresight knobs.* The maximum torque required to rotate the azimuth and elevation boresight knobs shall not exceed 5 pound-inches at standard ambient temperature and 10 pound-inches at the extreme temperatures specified in 3.6.3.

3.7.4.2 *Rheostat.* With the sight at the operating temperatures at 3.6.3, the torque required to rotate the rheostat knob shall not exceed 4 pounds.

3.7.5 *Electrical.* The continuity of the wiring shall be in conformance with the applicable wiring and continuity drawing requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for

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the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **General requirements.** Except where otherwise specified herein, the contractor's inspection system shall be in accordance with Specification MIL–I–45208.

4.3 *Initial production sample.* The contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein and in conformance with the contractor's approved inspection system.

4.3.1 *Initial production sample failure.* Should the production sample fail to pass the prescribed tests, the contractor shall make necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until corrective actions have been taken to eliminate deficiencies; and the production sample, as a result thereof, has been approved.

4.4 **Lot formation.** When inspection lot sizes and lot formation are applicable, they shall be in accordance with Standard MIL–STD–105 and shall also be defined in the contractor's inspection system.

4.4.1 *Defective units and lots.* Defective units and lots shall be processed as specified in Standard MIL–STD–105 and shall not be included as a part of the quantity specified in the contract until formal approval is obtained.

4.5 **Inspection provisions.** Components and subassemblies shall be examined in accordance with the inspection provisions contained in Specification MIL–F–13926 and 4.5.1 herein. Test and control of materials not covered by Supplementary Quality Assurance Provisions (SQAP) or Section 4 of Specification MIL–F–13926 shall be in accordance with "Test Data" provisions of Specification MIL–I–45208. Inspection related to Section 3 of this specification shall be performed in accordance with 4.5.2. The tabulated Classification of Defects listed herein shall constitute the minimum inspection to be performed by the contractor prior to Government acceptance. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.5.1 *Inspection of components.* Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP) 8635466. Components not covered by SQAP's shall be inspected in conformance with Specification MIL–F–13926.

4.5.1.1 *Contents of SQAP.* The contents of a SQAP represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of each component covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items.

4.5.1.2 *Contractor's inspection for all components not covered by SQAP.* Inspection for these components shall be in accordance with the inspection procedures of the inspection system required by Specification MIL–I–45208 and the sampling plan tables of Specification MIL–F–13926. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on

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the functional accuracy and performance of the sight. In addition, the list of Class 1 Defects in Specification MIL–F–13926 shall apply.

4.5.1.3 *Disposition of non-conforming product.* Disposition of non-conforming units shall be in accordance with Standard MIL–STD–105.

4.5.2 *Classification of defects and sampling plans.* Examination and testing to determine acceptance or rejection in compliance with the applicable requirements of Section 3 shall be conducted on a 100 percent inspection basis, initially. After conclusive evidence of the process average has been established in conformance with Standard MIL–STD–105 and the AQL's specified herein, the classification of defects and sampling plans specified in Table I, may be used in lieu of 100 percent inspection. The AQL value specified shall apply to each individual characteristic on a single defect basis.

4.5.2.1 *Disposition of reject lots.* Rejected lots shall be screened in conformance to 4.4.1 for all defective characteristics causing rejection of the lot. After the Government inspector has approved the corrective action taken, subsequent production lots may be submitted for acceptance in accordance with 4.5.2 and Standard MIL–STD–105.

TABLE I. *Classification of Defects*

Use Inspection Level II of Table III–A, with Sampling Plan Table IV–A of Standard MIL–STD–105.

Critical: None defined

Major:	*AQL 1.5% defective*	*Req't. Par*	*QA. Par.*	*Test Method*
101	Incorrect alinement	3.5	4.7.1	Standard Test Equipment (STE)
102	Incorrect knob travel	3.5.1	4.7.7.1	STE
103	Physical failure, post vibration	3.6.2	4.7.1.2	Standard Meas Equipment (SME)
104	Improper reticle retention	3.6.2.1	4.7.1.4	SME
105	Illumination	3.7.1	4.7.2	Visual
106	Improper focus	3.7.2	4.7.4	SME
107	Improper reticle diameter	3.7.3	4.7.3	STE
108	Improper torque, boresight knobs (Std Ambient Temp.).	3.7.4.1	4.7.5.2	SME
109	Improper torque, rheostat knob (Std Ambient Temp.).	3.7.4.2	4.7.5.1	SME
110	Improper continuity	3.7.5	4.7.6	SME

4.5.3 *Control test.* Three sights shall be selected at random from each 100 produced or from each months production whichever occurs first. Control samples shall be subjected to the examinations and tests specified herein to determine compliance with 3.2, 3.3 and 3.4. Three additional sights shall then be selected and all six samples shall be subjected to the test specified in 3.6. The examination and test shall be conducted by the contractor under surveillance of the Government representative at the contractor's facility or Government approved laboratory, using test equipment authorized for use by the contract or otherwise approved by the contracting officer.

4.5.3.1 *Control sample failure.* Should a control sample fail to meet the requirements specified in 4.5.3, the contractor shall cease those operations causing rejection until necessary corrections have been made by the contractor and approved by the Government. After approval of corrections and resumption of production, these requirements shall not be inspected on a control sample basis until at least 5 consecutive sights produced have been inspected to the performance requirements specified in 4.5.3 and accepted by the Government.

4.6 **Inspection equipment.** Inspection equipment shall conform to drawings listed in IEL 8635466, except as otherwise provided for under the alternative inspection procedure and inspection equipment requirements of Specification MIL–I–45208. Provisions of Specification MIL–I–45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in

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the Government approved *inspection system*. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.6.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.6.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 *Inspection equipment design requirements.*

4.6.2.1 *Device used to test the vibration.* The device shall conform to the "test facilities" specified in Specification MIL-F-13926.

4.6.2.2 *Device used to test temperature requirements.* This device shall conform to the test condition measurement requirements specified in Specification MIL-F-13926.

4.6.2.3 *Device used to test accessory requirement.* The device shall position the sight on a keyed locating surface with the mounting surface in a vertical plane and the key horizontal within one minute. The surface used for measuring the projected reticle image shall be graduated in one mil plus or minus 0.2 mil increments, horizontally and vertically in a manner suitable for measuring the subtended angle of the projected reticle pattern, and the boresight adjustment requirements. The surface shall be positioned 19° plus or minus one minute, from a horizontal plane. The equipment shall be capable of relocating any control within 0.05 mil of the original mounting position.

4.6.2.4 *Device to test boresight torque.* The device shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to ten pound-inches, the accuracy of which shall be within plus or minus one-quarter pound-inch through the entire zone of excursion. It shall be equipped with an adapter to accommodate the boresight knob.

4.6.2.5 *Device to test rheostat torque.* This device shall consist of a torque wrench type tester which meets the requirements specified in 4.6.2.4 and shall be equipped with an adapter to accommodate the rheostat knob.

4.6.2.6 *Device to test sealing requirements.* The device shall conform to the accuracy specified on Drawing F8565556.

4.6.2.7 *Device to test focus.* The device shall be a dioptometer of at least 6-power to determine the focus requirements of 3.7.2.

4.7 Test methods and procedures.

4.7.1 *Alignment.* The sight shall be positioned as specified in 3.5 when employing a testing device conforming to the requirements outlined in 4.6.2.3. When the line of sight has been adjusted to the position outlined in 3.5 the following tests shall be performed.

4.7.1.1 *Knob travel.* With the sight positioned as outlined in 4.7.1, the line of sight shall be displaced a minimum of ten mils in both the azimuth and elevation direction from the established position by rotating either the elevating or azimuth knobs or by rotating both knobs simultaneously. Return the line of sight to the position outlined in 4.7.1 and do not disturb either boresight knob setting until the tests in 4.7.1.2 and 4.7.1.3 have been completed.

4.7.1.2 *Vibration.* The sight shall be subjected to test Procedure V of Specification MIL-F-13926 except that the time specified in 3.6.2 shall apply. During this test do not disturb the boresight knob setting. The sight shall be visually inspected for conformance to the requirements of 3.6.2.

4.7.1.3 *Storage test.* This test will be performed by placing the sight in an acceptable type cold box (4.6.2.2) and gradually chilling

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the sight to the temperature of 3.6.1. The sight shall then be returned to room temperature and then placed in an acceptable type oven (4.6.2.2) with the temperature gradually increased to the temperature in 3.6.1. Be careful to avoid disturbing the knobs positions on the sight and perform a visual examination to determine if any damage or physical failure is evident.

4.7.1.4 *Reticle retention.* This test shall be performed after being subjected to the vibration and temperature tests specified above. The actual test is performed by remounting the sight on the testing device as outlined in 4.7.1 and observing where the line of sight falls on the target. The position of the line of sight shall be identical to the original position specified in 4.7.1 within the tolerance specified in 3.6.2.1.

4.7.1.5 *Operability.* At the conclusion of the tests outlined in 4.7.1.2, 4.7.1.3 and 4.7.1.4, the sight shall be subjected to the remaining tests in this specification.

4.7.2 *Illumination.* Energize the reticle lamp with the voltage specified in 3.7.1. The reticle shall be visually observed and appear clearly defined. Rotation of the rheostat knob shall cause the illumination of the reticle to increase.

4.7.3 *Reticle.* With the sight positioned as specified in 4.7.1 (i.e. with horizontal and vertical target lines in coincidence with the horizontal and vertical diameter of the reticle circle), observe the angular displacement within this inner diameter of the light source reticle at the above mentioned diameters. This angular displacement must conform to the requirement specified in 3.7.3.

4.7.4 *Focus.* This test will be performed using a calibrated dioptometer calibrated and adjusted as outlined below.

4.7.4.1 *Focusing the eyepiece.* Focus the dioptometer to obtain the sharpest image of its reticle.

4.7.4.1.1 *Target.* Direct the dioptometer line of sight toward any target positioned at or beyond the parallax distance, and select a well defined point on the target.

4.7.4.1.2 *Parallax.* Adjust the dioptometer's objective lens until all parallax has been removed. Parallax will be removed when there is no movement between the selected target point and a corresponding point on the dioptometer reticle. When parallax has been removed set the diopter scale to zero.

4.7.4.2 *Performing the test.* With the reticle in the sight illuminated coincide the dioptometer line of sight to that of the sight. Adjust the dioptometer objective assembly to obtain the sharpest image of the sight reticle. When this has been obtained, observe the diopter setting. This setting must conform to the requirement specified in 3.7.2.

4.7.5 *Torque.*

4.7.5.1 *Rheostat knobs.* The torque required to rotate the rheostat knob shall be checked with appropriate adapter and testing device conforming to the accuracy specified in 4.6.2.5. The initial torque required to rotate the rheostat shall be disregarded. The actual running torque read on the testing device shall conform to the requirement of 3.7.4.2.

4.7.5.2 *Boresight knobs.* The torque required to rotate the boresight knobs shall be checked with the appropriate adapter and testing device conforming to the accuracy specified in 4.6.2.5. The initial torque required to rotate the boresight knobs shall be disregarded. The actual running torque read on the testing device shall conform to the requirement of 3.7.4.2.

4.7.6 *Continuity.* Continuity shall be determined by means of a common continuity checker employing standard inspection techniques.

4.7.7 *Operating temperatures.* This test will be performed as outlined in Procedure II of Specification MIL–F–13926 except that the temperatures specified in 3.6.3 shall apply. Subsequent to the temperature test the sight shall meet the requirement specified in 3.6.3.

4.7.8 *Sealing.* This test will be performed

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in accordance with the set-up and operating procedure specified on Drawing F8565556.

4.7.9 *Packaging, packing, and marking.* Quality assurance provisions for packaging, packing, and marking shall be as specified in Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8635466. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Infinity Sight is primarily intended to project a reticle and control intensity of same, as projected into the gunner's periscope on an M60 Tank.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Provisions for initial production sample, including date of delivery (see 3.1).

6.3 Definitions.

6.3.1 *Special test equipment. (STE).* Special testing equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.3.2 *Standard measuring equipment (SME).* Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, and so forth.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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